

# Sello and AIR: two layers of evidence

## CURRENT VITA-FL IMPLEMENTATION

Sello checks the tool interaction. AIR binds the numeric inference.

### **Sello** Tool interaction

- 1 An authorised agent may call**  
mTLS + token: identity, Inference TEE, scope, certificate, expiry; job ownership.
- 2 A signed record of the tool call**  
Inference TEE signs action, input/output hashes, token reference and result status.
- 3 The receipt body stays encrypted**  
HPKE seals the Sello receipt for the owner.

### **AIR** Attested Inference Receipt

- 1 Model, input and result stay linked**  
Ed25519 signs their hashes and the quote hash.
- 2 The receipt key is bound to the quote**  
REPORTDATA: key + manifest + request.  
The AIR signature binds the result.
- 3 Deployment policy is checked**  
Replay RTMR3; compare Compose, image digest, chain, contracts and RPC.

#### SHARED LOG

Inference TEE logs Sello before reply



Agent verifies Transparency Log  
and logs AIR

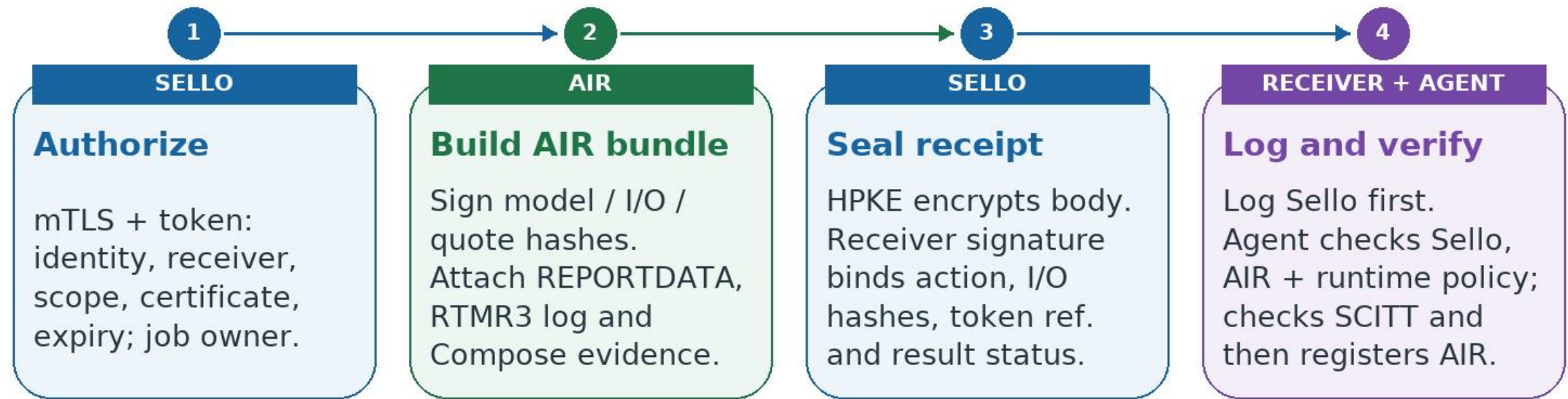
Sello-inspired profile · Figuera (2026)

AIR-style profile · Tsyrulnikov, Internet-Draft 02 (2026)

# Sello + AIR: checks along the inference workflow

## CURRENT VITA-FL IMPLEMENTATION

Sello authorizes and records the tool call; AIR binds the numeric inference evidence.



### Release rule

A verified result is returned only after receipt, AIR and log checks succeed.

Scope: per-inference Intel quote verification is absent; TEE trust relies on prior worker admission.

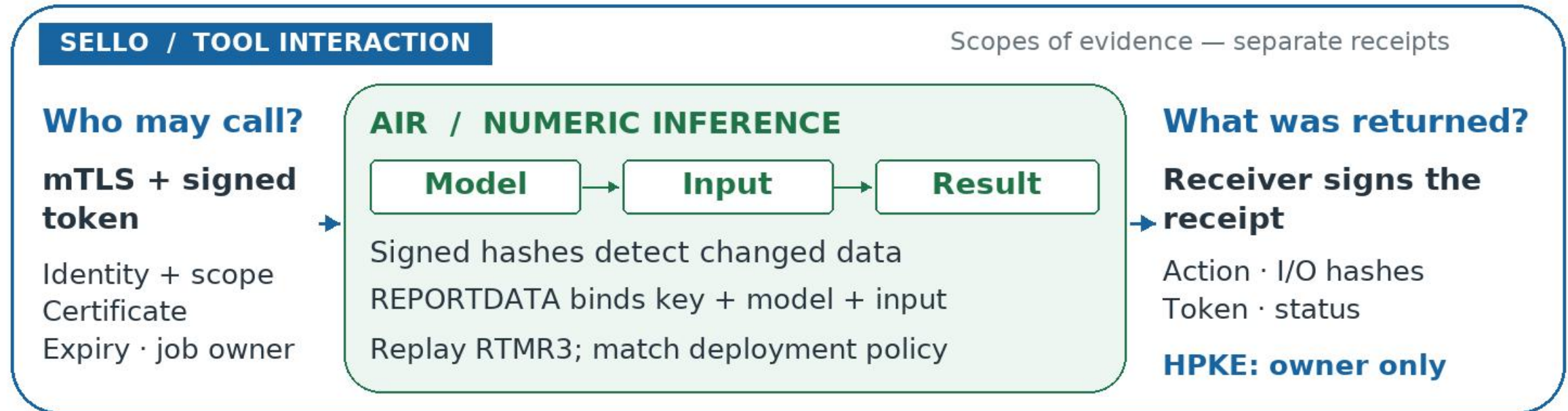
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# Two scopes of evidence, one verified tool result

## CURRENT VITA-FL IMPLEMENTATION

Sello covers the tool call; AIR binds its model, input and result.



### Log

Sello before reply → agent checks SCITT → AIR registered separately  
HPKE protects the Sello receipt only; the AIR bundle includes plaintext input/output.

Scope: per-inference Intel quote verification is absent; TEE trust relies on prior worker admission.

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# Sello + AIR: which changes are detected?

## CURRENT VITA-FL IMPLEMENTATION

Sello verifies the tool interaction; AIR binds the numeric inference evidence.

| SCENARIO                       | IMPLEMENTED CHECK                                                                        | ENFORCED OUTCOME                                  |
|--------------------------------|------------------------------------------------------------------------------------------|---------------------------------------------------|
| <b>Unauthorized caller</b>     | Sello · mTLS + signed token: subject, audience, scope, certificate thumbprint and expiry | Rejects identity, scope or token mismatches       |
| <b>Substituted tool result</b> | Sello · receiver signature: action, exact I/O, token reference and result status         | Detects substituted tool input or output          |
| <b>Exposed receipt body</b>    | Sello · HPKE encryption to the owner's key                                               | Hides the Sello body; AIR bundle is plaintext     |
| <b>Swapped inference data</b>  | AIR · signed data / quote hashes; REPORTDATA binds AIR key, manifest and request         | Detects data and quote-binding mismatches         |
| <b>Different deployment</b>    | AIR · RTMR3 replay; Compose / image; chain, storage, registry and RPC checks             | Rejects configuration outside the expected policy |
| <b>Missing log entry</b>       | SCITT · publication receipts verified before a successful verified tool result           | Withholds success; execution is not rolled back   |

Scope: per-inference Intel quote verification is absent; TEE trust relies on prior worker admission.

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